carb_coefs.cbn (deprecated)

Removed

`carb_coefs.cbn` was removed. The keyword-style reader (`carbon_coef_read.f90`) and the dead companion file `basins_carbon.tes` have also been deleted. The 22 daily hard-coded coefficient overrides that previously lived in `cbn_zhang2.f90` are gone; user-supplied values now flow end-to-end.

What replaces it

Carbon inputs have been split into two flat-table files that follow the same layout convention as the rest of the SWAT+ inputs (title line, header line, data rows):

- `carbon.bsn`: 28 basin-wide scalar parameters in a single data row. Initial pool fractions, water coupling, manure partitioning, temperature and water factor selectors, the `t_cbn_*` triple, biomixing and tillage curves, photodegradation, residue C:N / C:P controls, and the slow-humus initialization method (`mathers_method`).
- `carbon_lyr.bsn` — per-layer rate constants and CO2 allocation fractions. One row per layer group (1 = top, 2 = subsurface).

Both files are required when `bsn_cc%cswat == 2` (the dynamic carbon model; the activation value moved from 1 to 2). Either missing or malformed file triggers `error stop`; there is no silent fallback to compiled defaults.

Migration

Old keyword in `carb_coefs.cbn`	New location
`cbn_diagnostics`	retired; output gating now lives in `print.prt` via the `hru_cb_*` and `lsu_cb_*` rows
`hp_rate`, `hs_rate`, `microb_rate`, `meta_rate`, `str_rate`, `microb_top_rate`, `hs_hp`	per-layer rows of `carbon_lyr.bsn`
`a1co2`, `asco2`, `apco2`, `abco2`	per-layer rows of `carbon_lyr.bsn`
`org_frac` (`frac_seq`, `frac_hum_microb`, `frac_hum_slow`, `frac_hum_passive`)	`init_seq`, `init_microb`, `init_slow`, `init_passive` columns of `carbon.bsn`
`prmt_21`	`koc_c` column of `carbon.bsn`
`prmt_44`	`solc_ratio` column of `carbon.bsn`
`till_eff_days`	`till_eff_days` column of `carbon.bsn`
`rtof`	`manure_c_frac` column of `carbon.bsn`
`cbn_consolidation_factors` (2 values)	`bio_consol`, `till_consol` columns of `carbon.bsn`
`cbn_factor_approaches` (2 ints)	`tmpf_eqn`, `watf_eqn` columns of `carbon.bsn`
`tn`, `top`, `tx`	`t_cbn_min`, `t_cbn_opt`, `t_cbn_max` columns of `carbon.bsn`
`zz_bmix_coefs` (3 values)	`bmix_a`, `bmix_b`, `bmix_c` columns of `carbon.bsn`. `zz_` prefix dropped to match the column name.
`zz_emix_coefs` (3 values)	`tillmix_a`, `tillmix_b`, `tillmix_c` columns of `carbon.bsn`
`photo_degrade_factor`	`sfc_rsd_photodeg` column of `carbon.bsn`
`nmbr_soil_test_layers`, `soil_test`	retired; no replacement (sub-table was unused downstream)

Five new tunables were promoted from hard-coded constants and added to `carbon.bsn`:

- `n_act_frac` (formerly the `nactfr` local in `nut_nminrl.f90`)
- `cnr_cap`, `cnr_ref` (formerly the 500 and 25 hard-coded values in `cbn_surfrsd_decomp.f90`)
- `cpr_cap`, `cpr_ref` (formerly the 5000 and 200 hard-coded values in `cbn_surfrsd_decomp.f90`)

Related files

- `carbon.bsn`. Replacement for the basin-wide scalars.
- `carbon_lyr.bsn`. Replacement for the per-layer table.
- `codes.bsn`. The `cswat` switch. `cswat == 2` and `cswat == 3` are no longer accepted; set `0` (no carbon) or `1` (active carbon).
- Carbon outputs. Replacement for the `cbn_diagnostics` flag.